

أسئلة سنوات على آخر خضت يجازي :-

① The most accurate experiment to measure a resistor is :-

- a. Ohm's law exp.    b. Wheatstone bridge exp.    c. RC-circuit exp.

الجواب هو "b"

② To convert the Galvanometer to ammeter we connect :-

- a. Large resistor in series    b. Large resistor in parallel  
c. small resistor in parallel    d. small resistor in series

الجواب هو "c"

③ If the Potential difference across some resistor is doubled, then :-

- a. The resistance must be doubled    b. The current must be doubled  
c. The current must be halved    d. The resistance must be halved

الجواب هو "b"

④ In charging capacitor. To increase the time of charging we must increase the value of :-

- a. The capacitance and/or resistance    b. The capacitance (C)  
c. The electromotive force of the power supply    d. The resistance (R)

الجواب هو "a"

سارة العوس

⑤ The most accurate experiments to measure a large resistor is:-  
 a. Ohms Law exp. b. potentiometer exp. c. RC-circuit exp

الجواب هو "C"

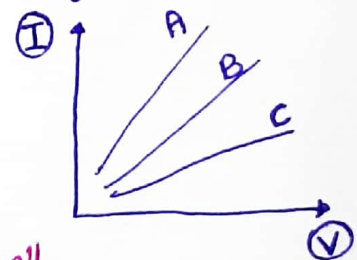
⑥ The main aim of Potentiometer experiment is:-

- a. test an old batteries      b. design a voltage divider  
 c. increase the efficiency of the instruments      d. measure the resistance with high accuracy

الجواب هو "d"

⑦ In ohm's law exp. a student obtained the result shown in the figure for three trials: single resistor, two resistor in series and two resistors in parallel, which of the following is true:-

- a. A  $\Rightarrow$  series, B  $\Rightarrow$  single, C  $\Rightarrow$  parallel  
 b. C  $\Rightarrow$  series, B  $\Rightarrow$  single, A  $\Rightarrow$  parallel  
 c. A  $\Rightarrow$  series, B  $\Rightarrow$  parallel, C  $\Rightarrow$  single  
 d. A  $\Rightarrow$  single, B  $\Rightarrow$  series, C  $\Rightarrow$  parallel



الجواب هو "b"

⑧ To convert a Galvanometer of internal resistance  $R_G = 120 \Omega$  into voltmeter. A large resistance of  $20 k\Omega$  connected in series with the Galvanometer. what is the maximum voltage in (V) can be measured by the constructed voltmeter. assuming that the maximum current sustained by the Galvanometer is equal  $0.4 mA$ ?

- A. 5 V      b. 20 V      c. 8 V      d. 12 V

$$V_{max} = I_{G, max} (R_G + R_x) \Rightarrow V_{max} = 0.4 \times 10^{-3} (120 + 20 \times 10^3)$$

$$V_{max} = 8 \text{ V}$$

الجواب هو "c"

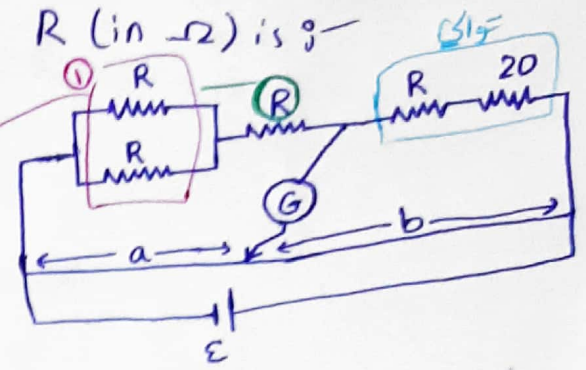


⑨ In the circuit shown, if the galvanometer reads zero at  $a = 12 \text{ cm}$  and  $b = 25 \text{ cm}$ . The value of  $R$  (in  $\Omega$ ) is —

- a. 9.4    b. 20    c. 10    d. 40

$$\frac{1}{R} = \frac{1}{R} = \left(\frac{R}{2}\right) \quad \text{مالتوازي}$$

$$\frac{R}{2} + R = \frac{3R}{2} \quad \text{مالتوالي}$$



$$\frac{\frac{3R}{2}}{R+20} = \frac{12}{25} \Rightarrow \frac{3R}{2} \times \frac{12R+240}{25}$$

$$75R = 24R + 480 \Rightarrow R = 9.4$$

الجواب هو "A"

ساعة العوض

⑩ A total resistance of  $3.0 \Omega$  is to be produced by combining two resistor,  $R$  and  $12 \Omega$  resistor. What is the value of  $R$  and how is it to be connected to the  $12 \Omega$  resistor?

- a.  $4 \Omega$ , series    b.  $4 \Omega$ , Parallel    c.  $2.4 \Omega$ , Parallel    d.  $2.4 \Omega$ , series

$$\frac{1}{12} + \frac{1}{R} = \frac{1}{3} \Rightarrow \frac{12+R}{12R} \times \frac{1}{3} \Rightarrow 12+R = 4R$$

$$3R = 12 \Rightarrow R = 4$$

( $R$  and  $12 \Omega$ ) connected in Parallel

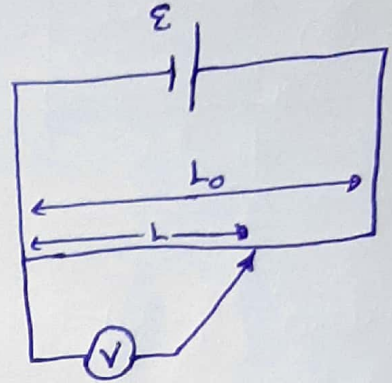
الجواب هو "b"

11) A Potentiometer was constructed as in figure shown, when the pointer was at the mid of the given resistor, the voltmeter read 7V. what is the voltage of the battery  $\mathcal{E}$ .

- a. 7V    b. 3.5V    c. 6V    d. 14V

$$V = \mathcal{E}$$

"A" هو الجواب



12) In discharging capacitor, if the voltage on it is halved after one minute. What is the value of the capacitance if it connect with  $70\text{ k}\Omega$ ?

- a. 0.9mf    b. 830F    c. 1.2 mf    d. 48.5 f

لاره القوس

$$R = 70\text{ k}\Omega = 70 \times 10^3 \Omega \quad / \quad t = 60 \text{ sec}$$

$$V = \frac{V_0}{2}$$

$$\frac{Q(t)}{C} = \frac{Q_0}{C} e^{-\frac{t}{RC}}$$

$$V = V_{\max} e^{-\frac{t}{RC}}$$

$$\frac{V_0}{2} = V_0 e^{-\frac{60}{70 \times 10^3 C}} \Rightarrow \frac{1}{2} = e^{-\frac{6}{7 \times 10^3 C}} \Rightarrow \ln \frac{1}{2} = -\frac{6}{7 \times 10^3 C}$$

$$\ln 2 = \frac{6}{7 \times 10^3 C} \Rightarrow C = \frac{6}{7 \times 10^3 \times \ln 2}$$

$$C = 1.2 \times 10^{-3} \text{ f} \Rightarrow \boxed{1.2 \text{ mf}}$$

"C" هو الجواب



(13) To convert a Galvanometer of internal resistance  $R_G = 120 \Omega$  into ammeter a small resistance connected in parallel with the Galvanometer using wire of resistivity equal  $(300 \Omega/\text{km})$ . If the maximum current can be measured by the constructed Ammeter is  $3\text{A}$ , assuming the maximum current sustained by the Galvanometer is equal  $0.2\text{mA}$ . What is the length of the wire used to build up this ammeter?

- a.  $2.67\text{cm}$     b.  $4\text{cm}$     c.  $3.5\text{cm}$     d.  $1.2\text{cm}$

$$R_s = \frac{I_{G, \max} R_G}{I_{\max} - I_{G, \max}} = \frac{(0.2 \times 10^{-3}) \times (120)}{(3) - (0.2 \times 10^{-3})} = 0.008 \Omega$$

$$L = \frac{R_s}{\lambda} \times 10^5 \text{ cm} = \frac{0.008}{300} \times 10^5 = 2.67 \text{ cm}$$

ل  
الطول

الجواب هو "a"

(14) In the "Wheatstone Bridge" exp. a student wanted to measure unknown resistance  $R$ . He found that  $I_g$  equal zero in the following state. What is the value of  $R$ ?

$$\frac{1}{2R} + \frac{1}{2R} = \frac{2}{2R} \Rightarrow R$$

$$3R + R = 4R$$

$$b \leftarrow (80) = 20 - 100 = \text{المقاومة المجهولة}$$

$$\frac{4R}{20} = \frac{80}{80} \Rightarrow R = 20 \Omega$$

